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A Comprehensive Study of Knowledge Editing for Large Language Models

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Abstract

Large Language Models (LLMs) have shown extraordinary capabilities in understanding and generating text that closely mirrors human communication. However, a primary limitation lies in the significant computational demands during training, arising from their extensive parameterization. This challenge is further intensified by the dynamic nature of the world, necessitating frequent updates to LLMs to correct outdated information or integrate new knowledge, thereby ensuring their continued relevance. Note that many applications demand continual model adjustments post-training to address deficiencies or undesirable behaviors. There is an increasing interest in efficient, lightweight methods for on-the-fly model modifications. To this end, recent years have seen a burgeoning in the techniques of **knowledge editing** for LLMs, which aim to *efficiently* modify LLMs' behaviors within specific domains while preserving overall performance across various inputs. In this paper, we first define the knowledge editing problem and then provide a comprehensive review of cutting-edge approaches. Drawing inspiration from educational and cognitive research theories [1–3], we propose a unified categorization criterion that classifies knowledge editing methods into three groups: *resorting to external knowledge*, *merging knowledge into the model*, and *editing intrinsic knowledge*. Furthermore, we introduce a new benchmark, **KnowEdit**, for a comprehensive empirical evaluation of representative knowledge editing approaches. Additionally, we provide an in-depth analysis of knowledge location, which can give a deeper understanding of the knowledge structures inherent within LLMs. Initially conceived as a means to steer LLMs efficiently, we hope that insights gained from knowledge editing research could shed light on the underlying knowledge mechanisms of LLMs. To facilitate future research, we have released an open-source framework, EasyEdit[1], which will enable practitioners to efficiently and flexibly implement knowledge editing for LLMs. Finally, we discuss several potential applications of knowledge editing, outlining its broad and impactful implications.

***Keywords*—** natural language processing, large language models, knowledge editing

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[1]https://github.com/zjunlp/EasyEdit.

The contributions of the authors are detailed in §CONTRIBUTIONS.

Ongoing work (v5): update Table 4 and add new references.

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1 Introduction

Knowledge is a fundamental component of human intelligence and civilization [4]. Its systematic structure empowers us to represent tangible entities or delineate principles through symbolic means, offering the capability to facilitate the articulation of intricate behaviors or tasks [5–7]. Throughout our lives, we humans continuously gather an extensive wealth of knowledge and learn to adaptively apply it in various contexts. The enduring exploration of the nature of knowledge and the processes by which we acquire, retain, and interpret it, continues to captivate scientists, which is not just a technical pursuit but a journey towards mirroring the nuanced complexities of human cognition, communication and intelligence [8–12].

Recently, Large Language Models (LLMs) like GPT-4 [13] have showcased a remarkable ability in Natural Language Processing (NLP) to retain a vast amount of knowledge, arguably surpassing human capacity [14–31]. This achievement can be attributed to the way LLMs process and compress huge amounts of data [32–35], potentially forming more concise, coherent, and interpretable models of the underlying generative processes, essentially creating a kind of "world model" [36–38]. For example, Dai et al. [39] have introduced the Knowledge Neuron (KN) thesis, which proposes that language models function similarly to key-value memories. Here, the multi-layer perceptron (MLP) weights in the core region [40] may play a crucial role in recalling facts from the training corpus, suggesting a more structured and retrievable form of knowledge storage within LLMs [41, 42]. Further insights come from the ability of LLMs to understand and manipulate complex strategic environments, whereas Li et al. [43] has demonstrated that transformers trained for next-token prediction in board games such as Othello develop explicit representations of the game's state. Patel and Pavlick [44] have revealed that LLMs can track boolean states of subjects within given contexts and learn representations that reflect perceptual, symbolic concepts [36, 45–47]. This dual capability indicates that LLMs can serve as extensive knowledge bases [48–59], not only storing vast amounts of information but also structuring it in ways that may mirror human cognitive processes.

However, LLMs have limitations like factual fallacy, potential generation of harmful content, and outdated knowledge due to their training cut-off [60–63]. Retraining to correct these issues is both costly and time-consuming [64–68]. To address this, recent years have seen a surge in the development of knowledge editing techniques specifically tailored for LLMs, which allows for cost-effective post-hoc modifications to models [69–71]. This technique focuses on specific areas for adjustment without compromising overall performance and can help understand how LLMs represent and process information, which is crucial for ensuring the fairness, and safety in Artificial Intelligence (AI) applications [72–76].

This paper first attempts to provide a comprehensive study of the development and recent advances in knowledge editing for LLMs. We first introduce the architecture of Transformers, mechanism of knowledge storage in LLMs (§2.1), and related techniques including parameter-efficient fine-tuning, knowledge augmentation, continue learning and machine unlearning (§2.2). Then we introduce preliminary (§3.1), formally describe the knowledge editing problem (§3.2), and propose a new taxonomy (§3.3) to provide a unified view on knowledge editing methods based on the educational and cognitive research theories [1–3]. Specifically, we categorize knowledge editing for LLMs into: resorting to external knowledge (§3.3.1), merging knowledge into the model (§3.3.2), and editing intrinsic knowledge (§3.3.3) approaches. Our categorization criterion is summarized as follows:

- **Resorting to External Knowledge.** This kind of approach is similar to the recognition phase in human cognitive processes, which needs to be exposed to new knowledge within a relevant context, just as people first encounter new information. For example, providing sentences that illustrate a factual update as a demonstration of the model allows initial recognition of the knowledge to be edited.
- **Merging Knowledge into the Model.** This kind of approach closely resembles the association phrase in human cognitive processes, in which connections are formed between the new knowledge and existing knowledge in the model. Methods would combine or substitute the output or intermediate output with a learned knowledge representation.
- **Editing Intrinsic Knowledge.** This approach to knowledge editing is akin to the mastery phase in human cognitive processes. It involves the model fully integrating knowledge into its parameters by modifying the weights and utilizing them reliably.